Teaching and Teacher Education 132 (2023) 104247

Contents lists available at ScienceDirect

Teaching and Teacher Education

journal homepage: www.elsevier.com/locate/tate

Research paper

Math is cool – for school? Insights into preservice teachers' views on the mathematics discipline between an academic and a job-related focus

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HIGHLIGHTS

- Preservice mathematics teachers view the mathematics discipline with both an academic and a job-related focus.
- Views indicate low, moderate or high affinity rather than a pronounced academic or job-related focus.
- Preservice teachers showing low affinity tend to see the content as lacking relevance for teaching.
- Affinity toward the discipline of mathematics is independent of background variables.
- Reading quotes on others' views of mathematics did not prime preservice teachers' focus.

ARTICLE INFO

Article history:
Received 11 November 2022
Received in revised form
19 June 2023
Accepted 24 June 2023
Available online 03 July 2023

Keywords:
Preservice teachers
Content
Mathematics discipline
Views
Job-related focus
Higher education

ABSTRACT

In academic teacher education, preservice teachers may develop complex views on their chosen scientific discipline of study, as a tension may exist between a purely academic focus and a job-related focus. This study reports quantitative data from $N = 259$ preservice mathematics teachers indicating their views on the mathematics discipline in four facets of task value and enjoyment of mathematics as well as in an open statement. Cluster analysis of the five quantitative variables suggests three disjunct clusters, and qualitative data support the cluster structure. Data are discussed regarding higher education and the role of discipline content in teacher education.

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1. Introduction

Becoming a teacher is an academic endeavor. In many countries, secondary school teachers need specialization in scientific disciplines underlying their teaching subjects (Wasserman et al., 2018). To provide preservice teachers this specialization, universities, as venues of academia and science, aim to educate preservice teachers so they become experts of the studied disciplines (Code et al., 2016). In this education, preservice teachers should develop a personal relevance of the discipline they are studying; that is, they should have an appreciation for and views on the discipline (Warwick & Ottewill, 2004). Especially for preservice mathematics teachers, a tension might arise between focusing on mathematics as a pure scientific and academic discipline and focusing on how mathematics is relevant for their future teaching activities (Afdal, 2017). In mathematics, only loose ties seem to connect the application-oriented mathematics taught at school and the mathematics discipline at university with a truly scientific body (Geisler & Rolka, 2021; Moreira & David, 2008; Ufer et al., 2017). While a job-related focus may highlight the relevance of mathematics for preservice teachers, such a focus might be perceived as artificial, since the mathematics discipline cannot always be transferred immediately into teaching content at school. By contrast, an academic focus might let preservice teachers understand the value of science and academia, but such a focus may also be perceived as "nerdy" or not suitable for preservice teachers. Given these seemingly contrasting

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https://doi.org/10.1016/j.tate.2023.104247
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focuses, this paper aims to understand how preservice teachers orient in this tension between an academic and a job-related focus on mathematics as a discipline by investigating their views on mathematics. By doing this, the construct of task value is consulted as theoretical framework to capture the broad range of preservice teachers' views.

1.1. Preservice teachers' job-related focus on the mathematics discipline

In general, preservice teachers are perceived to be rather focused on their future job and not on the studies of a particular discipline at university (Göller & Besser, 2021; Malmberg, 2006). Thus, many preservice teachers appreciate understanding how studying mathematics is relevant for their future teaching activities (Koponen et al., 2016) or are more interested in mathematics that is related to school (Ufer et al., 2017). Although, there might be differences between students from different teacher education programs (Retelsdorf et al., 2014), the question how discipline content is relevant for the future job is prevailing in preservice mathematics teachers' views on the mathematics discipline (Gildehaus & Liebendörfer, 2021; Zazkis & Leikin, 2010).

To better understand how preservice teachers view the mathematics discipline as relevant for the future job, the role of the mathematics discipline in teacher education may be consulted first. Offering knowledge for a direct transfer to teaching contexts is not the main purpose why teacher education includes scientific disciplines (Korthagen, 2010). Research suggests that teachers' professional knowledge is built on knowledge deriving from studying disciplines (Ball et al., 2008; Ernest, 1989; Shulman, 1986). Nevertheless, most scholars agree that discipline content cannot be directly applied to everyday teaching (Ball et al., 2008; Risan, 2020; Sjølie, 2014). While content from educational sciences may be more or less directly applied at school, preservice teachers need to study mathematics as discipline behind the teaching subject as a discipline that is rather seldom connected to teaching at school, alike other science, technology, engineering, and mathematics (STEM) subjects (Moreira & David, 2008). For example, preservice mathematics teachers may learn about complex analysis that is not part of the mathematics curriculum at school. Taken together, preservice teacher may not grasp the role of the mathematics discipline in teacher education directly, and, thus, not all preservice teachers agree that studying the mathematics discipline is useful for becoming a teacher (Koponen et al., 2016).

Providing a job-related focus on the mathematics discipline is a common attempt in teacher education to introduce preservice teachers to the discipline of mathematics. For example, one way to provide a job-related focus on the mathematics discipline is to interweave courses from the mathematics discipline with internships (Cardetti & Truxaw, 2014; Philipp et al., 2007), or ask preservice teachers to reflect on how content can be used in particular teaching settings (Massolt & Borowski, 2018; Wasserman et al., 2017; Weber et al., 2020). Those field studies show that such interwoven courses lead preservice teachers to find the mathematics discipline more valuable and relevant (Beswick, 2006; Eichler & Isaev, 2022; Hart, 2002) and to be more motivated (McGuffey et al., 2019). In higher education in general, communicating the content's relevance directly to students or letting them self-reflect on the relevance of content is an effective way to foster perceived utility of content, interest, and motivation (Acee & Weinstein, 2010; Harackiewicz & Priniski, 2018). Nevertheless, some studies also point out that such interventions need to be matched carefully to the learning group (Canning et al., 2019; Canning & Harackiewicz, 2015). Transferred to the context of teacher education, those are hints that addressing job-related relevance has a positive impact on preservice teachers' views. In sum, many preservice teachers hold a job-related focus on the mathematics discipline, and universities often address this focus thrivingly.

1.2. Preservice teachers' academic focus on the mathematics discipline

Albeit many preservice teachers hold a job-related focus on the mathematics discipline, not all preservice teachers grasp the connections between the mathematics discipline and future teaching activities (Auslander et al., 2019; Fischer, 2021). Alternatively, preservice teachers might view the mathematics discipline as a discipline itself, that is, with an academic focus. As such, preservice teachers may refer to studies of mathematics itself in their views on the mathematics discipline, for example by pointing out, in how far mathematics per se is part of their identity (Lahdenperä & Nieminen, 2020). Preservice teachers might highlight insights in how mathematics works and what role it plays for our society and our world (e.g., studying how algorithms work to truly understand search engine algorithms; Wood et al., 2012).

In general, sometimes it might be more appropriate for learners to understand why certain content is learned instead of understanding how the content is related to the learners' lives, as suggested by research on learning at schools (Brophy, 2008). Connections between the learning content and learners' lives might be perceived as artificial, at least from learners' point of view. This approach can be transferred to academic mathematics teacher education, as not all preservice teachers grasp how studying the mathematics discipline is useful for future teaching activities. Studying the mathematics discipline may be valuable for preservice teachers to arrive an in-depth understanding, they aim to achieve in a topic and being fascinated by these (Albrecht & Karabenick, 2019). For example, conducting a proof of a theorem deepens one's own mathematical knowledge and logical reasoning skills and, in addition, can be fascinating due to the "eureka" moment (Liljedahl, 2005). Furthermore, the mathematics discipline may be valuable for its aesthetic nature (Brophy, 2008); for example, a clean proof may be more aesthetic than obtuse calculations (Meyer & Eley, 1999). These reasons for appreciating the mathematics discipline involve personal experiences that are likely to result in positive affection (Liljedahl, 2005).

Besides those theoretical considerations, studies on preservice teachers' orientations suggest that for some preservice teachers, the discipline studied is more interesting than becoming a teacher (Dörrenbächer-Ulrich et al., 2019; Göller & Besser, 2021). However, such academic focus does not match the dominant orientation of preservice teachers, that is, an orientation toward their future job.

1.3. The multidimensionality of preservice teachers' views on the mathematics discipline

Although the two focuses on the mathematics discipline described above (i.e., an academic focus and a job-related focus) relate to different aspects in preservice teachers' views, they might not be mutually exclusive. Academic teacher education can address these individual focuses and their combination to let preservice teachers better understand the role of discipline content (Risan, 2020). Most research on this subject has focused on preservice teachers holding a job-related focus on the mathematics discipline and/or how to positively influence their motivation and views using a job-related focus. However, a research gap exists on the relationship between an academic focus and a job-related focus on the mathematics discipline. Some insights suggest that most preservice teachers hold some of both focuses in their views, while some

preservice teachers are solely focused on how studying discipline content is related to their future job (Billich-Knapp et al., 2012; Lorentzen et al., 2019; Quinlan & Salmen, 2019). This may change over the duration of studies, as preservice teachers studying in their masters' degree can understand more elaborately how an academic and a job-related focus on discipline content go hand-in-hand (Hermanns & Keller, 2021). Other studies suggest that instead of having either a strong academic focus or job-related focus, some preservice teachers rather hold an overall appreciation of the discipline studied that is high or low, independent their dominant focus (Dörrenbächer-Ulrich et al., 2019; Lee et al., 2022).

1.4. Understanding preservice teachers' views with the construct of task value

The literature on preservice teachers' views on the studied discipline as presented above does not follow a unified theoretical framework. Many of the previous studies used a qualitative approach or focus on specific aspects of views such as interest (Ufer et al., 2017). Since unified frameworks to investigate views on scientific and academic disciplines are spare (Kind et al., 2007), we refer to the construct of task value from Eccles and Wigfield's (1995) expectancy value theory as a theoretical framework to investigate preservice teachers' multidimensional views on the mathematics discipline. Task value captures separate aspects of how a task (i.e., studying the mathematics discipline) can be appreciated. Since preservice teachers' views on the mathematics discipline appear to be multidimensional as outlined above, such broad and differentiable framework as task value appeared worth applying to our context. In different contexts, the construct of task value and its subfacets proofed as a framework to capture views of preservice teachers aptly, for example motives of study choice (Pohlmann & Möller, 2010). Our aim was to transfer the construct of task value and its subfacets to the field between an academic focus and a job-related focus in preservice teachers' views on the mathematics discipline. Task value includes the enjoyment directly deriving from the task (intrinsic value), the personal importance of doing well in the task (attainment value), and the usefulness of the task for short- or long-term goals (utility value; Eccles & Wigfield, 2002, 2020). These considerations derive from research on different students and learners. Though the three aspects of task value (intrinsic value, attainment value, utility value) are often used as a framework, each study shifts a different focus on the investigated subfacets regarding the research context. For example, Egger and Ophoff (2020) found that for preservice teachers educational research can be useful for writing a thesis or for future teaching activities. In the following, we outline how preservice teachers can appreciate the mathematics discipline regarding the subfacets of task value given the tension between addressing an academic or a job-related focus and how such form of appreciation might be fostered.

Intrinsic value might be found when preservice teachers enjoy dealing with the mathematics discipline itself, for example, due to interest or enjoyment of its scientific nature. Preservice teachers who view studying the mathematics discipline as useful for future teaching activities may also show a high intrinsic value due to their interest of becoming a good teacher (Kunter et al., 2011; Priniski et al., 2018). Intrinsic value might be fostered when teaching addresses how enthusiastic other learners are about the momentary enjoyment of the task (Rosenzweig et al., 2021). Though definitions of intrinsic value cover a range of aspects in which intrinsic value might appear, measures often lack such range of aspects (Eccles & Wigfield, 2002; Gaspard, Dicke, Flunger, Schreier, et al., 2015). Especially in the context of the mathematics discipline, enjoyment might derive from the fascination of understanding the mathematics discipline and experiencing a "eureka" moment (Liljedahl, 2005). Thus, enjoyment of mathematics appears to be a substantial part how studying the mathematics discipline can be appreciated (Meyer & Eley, 1999); though, enjoyment of mathematics exceeds what is captured by task value. Although enjoyment per se is included in intrinsic value, enjoyment regarding the academic character of the learning content represents an own aspect of how the mathematics discipline can be appreciated (Brophy, 2008; Kunter et al., 2011).

Attainment value might be found when preservice teachers want to do well in the mathematics courses, either because the grade is important for them or the content and competences learned are important for them. The content of the mathematics discipline may be important as preservice teachers aim to become experts in mathematics or good teachers. This value can be fostered by highlighting that doing well in the task is important for the types of people the learners want to become (Johnson & Sinatra, 2013; Rosenzweig et al., 2021).

Appreciation of the mathematics discipline might also derive when preservice teachers study mathematics to understand a personal focal issue (Albrecht & Karabenick, 2019). In the context of teacher education, the mathematics discipline may indeed be such a focal issue, given that the preservice teachers chose to study mathematics.

Utility value might be found inter alia when preservice teachers think that studying the mathematics discipline will become useful in the future. This may be in the short-term future (that is, for future courses at university) or in the long-term future (that is, in the future job as a teacher; Gaspard, Dicke, Flunger, Schreier, et al., 2015). Thus, utility value might be found in either focus on the mathematics discipline preservice teachers hold – academic at university or job-related for teaching. Deriving from research on motivation in higher education, utility value interventions have been able to foster learners' perceived utility, motivation, and interest when the relevance of learning content is communicated (Harackiewicz & Priniski, 2018; Soicher & Becker-Blease, 2020). However, communicating a prescribed form of relevance does not work for every learner, since relevance needs to be closely aligned with learners' interests (Canning & Harackiewicz, 2015; Durik et al., 2015; Edwards et al., 2021; Liebendörfer & Schukajlow, 2020). Nevertheless, preservice teachers appear to be a rather homogenous group due to their collective career aim.

2. Present study

Given these considerations of how an academic and a job-related focus on the mathematics discipline look like and to what extends preservice teachers hold such focuses, we aimed to investigate the following research question: What views on the mathematics discipline do preservice teachers hold given the tension between an academic and a job-related focus? Besides gaining insights into an academic and a job-related focus in preservice teachers' views on the mathematics discipline, this study aimed to investigate the interaction of both focuses. Our main interest lay on how preservice teachers appreciate the mathematics discipline in different aspects regarding task value and enjoyment of mathematics. In detail, we scrutinized to what extend preservice teachers appreciate the mathematics discipline with an academic focus (i.e., utility for future university courses or enjoyment of mathematics per se) and to what extend preservice teachers appreciate the mathematics discipline with a job-related focus (i.e., utility for future teaching activities) as well as to what extend preservice teachers appreciate the mathematics discipline in general (i.e., intrinsic value and personal importance).

We originally planned to use quotes with an academic or a job-

related focus on studying mathematics to prime preservice teachers' views on the mathematics discipline in an experimental 2 × 2 design. We varied whether the quotes addressed an academic focus on studying mathematics (yes vs. no) and whether they addressed a job-related focus on studying mathematics (yes vs. no). The underlying hypotheses for this research question can be found in the preregistration available at OSF. Although in other studies reading quotes had an impact on learners' appreciation of mathematics (Gaspard, Dicke, Flunger, Brisson, et al., 2015; Rosenzweig et al., 2019), this intervention format did not lead to any group differences in our study, as briefly described in the results section below. Therefore, we assume that preservice teachers' appreciation of and views on the mathematics discipline are rather stable (Hart et al., 2019; Rach, 2023). Thus, we decided to reanalyze the whole dataset exploratorily following a cluster-building approach on the following research question: When focusing specifically on the tension between an academic and a job-related focus, what views on the mathematics discipline do preservice teachers hold?

3. Methods

We conducted an online experiment via Unipark (Tivian, 2022). All materials were presented in German and can be found, with the initial preregistration and the dataset, in the online supplementary material at OSF.

3.1. Sample

Participants were recruited from ten German universities with mathematics teacher education programs. All universities followed the same federal teacher education system where preservice teachers have to choose two major subjects for teaching (and a third eventually). Bachelor studies mainly consist of discipline content courses in the subjects and include only a few educational science elements, such as internships. After graduating with a Bachelor of Arts or a Bachelor of Science, preservice teachers need to study for a Master of Education degree, which focuses on didactics and educational sciences. Preservice teachers were recruited online via email by university lecturers and student organizations.

In total, $N = 307$ participants completed the survey; data from 36 participants were excluded from further analyses as they were not enrolled in a mathematics teacher program ($n = 27$), stated that they did not aim to become a teacher ($n = 7$), or did not give consent to use their data ($n = 2$). Furthermore, 7 participants were excluded due to a failed attention check, and 5 participants were excluded due do unreasonable duration of participating in the study; that is, they took a long break between being primed with the quotes and answering the questions on the dependent variables. Thus, our final sample consisted of $n = 259$ participants. Due to a technical issue, data about background information was missing for 61 participants, but we found no significant differences in the dependent variables regarding whether background data was missing or not (all $p \geq .178$). Of the 198 participants for whom background data were available, 117 (59.1%) were female, 3 (1.5%) diverse, and 5 (2.5%) did not specify their gender. Participants were between 18 and 35 years old ($M = 21.04$; $SD = 2.57$) and were studying mathematics in semester 2 to 20 ($M = 3.99$; $SD = 2.49$). As a second or third subject, 35.9% of the participants studied a science subject, 23.7% a language subject, 29.8% a social science subject, and 14.1% a subject not fitting in these categories (e.g., sports). A third subject was studied by 3.5% of the participants. The second subject was studied in semester 1 to 20 ($M = 4.09$; $SD = 2.54$). The third subject was studied in semester 2 to 4 ($M = 2.67$; $SD = 1.03$). Participants took on average 9:27 minutes ($SD = 4{:}51$) to complete the survey. Participation was reimbursed by a 5€ voucher for either an online bookstore or an online clothing store.

3.2. Procedure

First, participants gave their informed consent and stated whether they were studying in the requested degree program and whether they aimed to become to a teacher. The reason we gave as to why preservice teachers were asked to participate in the study was to capture their views on studying mathematics. Participants were randomly assigned to one of four experimental groups: *no quotes condition* (not addressing an academic focus and not addressing a job-related focus; $n = 66$), *only job-related focus condition* (not addressing an academic focus and addressing a job-related focus; $n = 62$), *only academic focus condition* (addressing an academic focus and not addressing a job-related focus; $n = 64$), and *mixed condition* (addressing an academic focus and addressing a job-related focus; $n = 67$).

We refer to the latter three conditions as *quotes conditions*. In the no quotes condition, participants did not get any treatment and were directly asked to indicate the dependent variables. Participants in the quotes conditions were asked to read six quotes about studying mathematics. Participants had to take at least 10 seconds to read each quote. As dependent variables, we assessed intrinsic value, personal importance, utility for university, utility for teaching, and enjoyment of mathematics on Likert scales and their views on the mathematics discipline in an open question for all participants. Participants assigned to one of the three quotes conditions were asked to indicate in a manipulation check whether at least one quote mentioned direct links to teaching activities. Furthermore, we debriefed participants that quotes were fictitious and asked them to comment on the quotes. In the end, all participants were asked to give consent to use their data and provide any comments on the study.

3.3. Materials

Participants in the three quotes conditions were asked to read six quotes before answering the survey. Each quote provided one perspective on what in-service mathematics teachers liked about studying mathematics. Quotes were 2–4 sentences long. Quotes in the only academic focus condition highlighted appreciation for the discipline of mathematics for its own sake. That is, quotes describe a fascination of the mathematics discipline due to its rational character and enjoyment of understanding and engaging in mathematics (e.g., "In my studies, I learned how different sub-disciplines of mathematics (for example, geometry and algebra in algebraic geometry) are related. I could see how the findings of one sub-discipline almost magically help in the other sub-discipline."). In the only job-related focus condition, quotes highlighted appreciation for mathematics due to its utility for teaching. That is, the quotes described gaining background knowledge of content and curriculum that helped explain and present mathematics at school (e.g., "For me, studying mathematics was beneficial for teaching. Thanks to my studies, it is easy for me to spontaneously find a suitable explanation when students have problems understanding."). Although in this condition some content from the mathematics discipline was mentioned, it was only mentioned to due to its utility for teaching. In the mixed condition, three quotes from the only academic focus condition and three quotes from the only job-related focus condition were used. Quotes were shown in alternating order, starting with a quote from the only academic focus condition.

3.4. Measures

3.4.1. Quantitative measures

For all task value and enjoyment of mathematics measures, instructions asked participants to focus on the study of the mathematics discipline. In addition, all items were adjusted in formulation for this aim. We unified scales to 6-point Likert scales with the same wording (1 = "do not agree at all" to 6 = "completely agree").

Task value. We captured four facets of task value. We used a subscale for intrinsic value (Gaspard, Dicke, Flunger, Schreier, et al., 2015) that captures the enjoyment of doing a task in 4 items, such as "Studying the mathematics discipline is fun to me.". The scale reached a Cronbach's α of 0.94.

For attainment value, we focused on personal importance, as this highlights the relations of mastering the task to the person's identity and not the importance of good grades. We used the scale from Gaspard and colleagues (2015) consisting of 6 items, such as "It is important to me to know a lot of the mathematics discipline.". We found a Cronbach's α of 0.91.

For utility value, we focused on both short- and long-term goals, in relation to utility for university and utility for teaching. We adapted the 2-item scale of Gaspard and colleagues (2015) capturing originally utility for school. We added 3 items to increase reliability, capturing additional aspects of utility value regarding university. These items were adjusted from a scale on personal or communal utility value of science (Shin et al., 2019). In total, our scale consists of 5 items, such as "The study of the mathematics discipline is valuable because it will help me in my further studies.". In total, the scale reached a Cronbach's α of .80.

To capture teaching-specific utility, we used the 3-item scale from Quast, Rubach, and Lazarides (2021), capturing originally utility of using digital media for teaching. Analogously to utility for university, we added 2 items from Shin et al. (2019). All items were adjusted to capture utility of studying the mathematics discipline for teaching. In total, the scale consists of 5 items, one example being "Studying the mathematics discipline has advantages for teachers.", and reached a Cronbach's α of 0.92.

As our study is the first to use this particular instrument, we conducted a confirmatory factor analysis with jamovi (The jamovi project, 2021) for the four factors of task value (intrinsic value, personal importance, utility for university, and utility for teaching). A four-factor structure was supported ($\chi^2 = 361$, $df = 164$, $p < .001$, CFI = 0.950, TLI = 0.942, RMSEA = 0.0680, SRMR = 0.0461).

Enjoyment of mathematics. In addition to task value, we assessed participants' subject-specific enjoyment of mathematics deriving from the rational character of mathematics. The scale of Meyer and Eley (1999) consists of 6 items, such as "I enjoy the rigor and precision of studying the mathematics discipline.". We found a Cronbach's α of 0.88.

3.4.2. Views on the mathematics discipline

Participants were asked to write a few sentences on their personal view of what role the mathematics discipline plays in their studies.

3.4.3. Background data and control measures

We collected information on participants' age, gender, second subject of study, third subject of study, and semester of study for each subject. Furthermore, participants were asked to indicate in dichotomous items whether they were studying in the requested degree program (mathematics for bachelor students to become a teacher) and whether they aimed to become a teacher.

3.4.4. Manipulation check

To control whether participants read and understood the quotes, we asked in the quotes conditions whether at least one quote mentioned direct links to teaching activities in a dichotomous item. Answers in the only academic focus condition differed significantly from answers in the only job-related focus condition ($\chi^2(1) = 74.75$; $p < .001$) as well as from answers in the only academic focus condition and the mixed condition ($\chi^2(1) = 60.50$; $p < .001$).

3.4.5. Comment on quotes

To get feedback on our research material, we asked participants in the quotes conditions in a voluntary open statement question to comment on the quotes they read. We coded whether they stated agreement, partial agreement, or disagreement with the quotes. There was no significant difference in the level of agreement between conditions ($\chi^2(4) = 7.03$; $p = .134$).

3.5. Analyses

All analyses were conducted in SPSS 28. To investigate their views, participants were clustered according to their responses to the five facets of appreciation of the mathematics discipline (intrinsic value, personal importance, utility for university, utility for teaching, and enjoyment of mathematics). Hierarchical clustering (Ward's method, squared Euclidean distance) was used to determine the number of clusters, and K-means clustering was used to identify the cluster membership. One-way ANOVAs with Tukey's post-hoc tests were used to investigate the differences between the means of the three clusters on each of the five factors. Background variables and qualitative answers were compared between clusters using ANOVAs (continuous data) and Pearson's χ^2 tests (categorical data).

For a more detailed analysis of participants' views on the mathematics discipline in their studies, we used qualitative content analysis (Mayring, 2015). We assumed that participants' views on discipline content could include multiple focuses, so each aspect was used as one coding unit. The coding manual codes whether participants used an academic focus or a job-related focus in their views on the mathematics discipline and whether these focuses were used for a positive or negative evaluation. An academic focus can be used to both positively and negatively evaluate the mathematics discipline due to its nature as a rational discipline. A job-related focus could be used in positive evaluations when the mathematics discipline was relevant for teaching and in negative evaluations when the mathematics discipline lacked relevance for teaching. Two coders showed a substantial interrater reliability (Cohen's $\kappa = 0.68$). Answers that were not elaborated enough were not coded. Categories are described as follows:

Positive academic focus. Participants' answers were coded in this category when they stated positive affection toward the mathematics discipline. These answers related directly to the discipline (e.g., "I am fascinated by mathematics") or to activities (e.g., "I enjoy puzzling over the tasks") or to characteristics related to the discipline, especially logic (e.g., "I like the scientific work").

Negative academic focus. Answers coded in this category indicated negative affection toward the mathematics discipline itself due to its complexity (e.g., "it causes me a lot of stress to deal with the mathematics discipline").

Positive job-related focus. This category codes all aspects referring to the usefulness or relevance of studying the mathematics discipline for teaching mathematics. The category included specific (e.g., advanced knowledge) or general arguments for why studying the mathematics discipline is perceived to be useful for participants' future job, e.g., "The mathematics discipline can help me

later as a teacher to explain certain topics better because I have more in-depth knowledge." However, even when studying mathematics was seen as useful for teaching, it was not seen as the most useful content at university for teaching.

Negative job-related focus for lacking relevance. Answers in this category included statements on the lacking or low relevance of studying the mathematics discipline for participants' future job. These statements entailed the desire for a stronger link between the mathematics discipline and participants' future job (e.g., "For me it is unnecessary because I want to be a teacher and not a mathematician") or the wish for other content (e.g., "it would help more to practice teaching"). Some participants highlighted that studying mathematics demands a lot of effort (e.g., time and stress) and related this effort to the usefulness (e.g., "the mathematics discipline courses take up an extremely large amount of time, which I find disproportionate compared to other parts of my studies").

4. Results

4.1. Influence of quotes on appreciation of the mathematics discipline

ANOVAs did not indicate significant main effects of addressing an academic focus (all $p \geq .174$) or a job-related focus (all $p \geq .157$) on intrinsic value, personal importance, utility for university, utility for teaching, and enjoyment of mathematics. The interaction effects of addressing both focuses were not significant on intrinsic value, utility for university, utility for teaching, and enjoyment of mathematics (all $p \geq .059$). The interaction effect on personal importance reached significance ($p = .014$; partial $\eta^2 = 0.023$). Experimental groups did not differ in age, gender, semester of study, or second or third subject of study (all $p \geq .545$). In consequence, we decided to include all participants for the cluster analysis, regardless the experimental group they had been initially assigned to.

4.2. Cluster analysis of participants' appreciation of the mathematics discipline

Participants were clustered according to their means for the five variables of appreciation of the mathematics discipline (intrinsic value, personal importance, utility for university, utility for teaching, and enjoyment of mathematics). Hierarchical clustering suggested three clusters. K-means clustering with number of clusters fixed to three was used to identify the cluster membership of each participant. Post-hoc ANOVAs indicated that the three clusters differed significantly in intrinsic value ($F(2,256) = 243.78$, $p < .001$), personal importance ($F(2,256) = 270.49$, $p < .001$), utility for university ($F(2,256) = 71.55$, $p < .001$), utility for teaching ($F(2,256) = 125.92$, $p < .001$), and enjoyment of mathematics ($F(2,256) = 244.49$, $p < .001$). Tuckey's post-hoc tests indicated significant differences between all three clusters in all five variables (all $p < .001$). The clusters can be described as follows.

Cluster 1. Participants in this cluster indicated values below the scale mean for intrinsic value ($M = 2.17$, $SD = 0.72$), personal importance ($M = 2.39$, $SD = 0.62$) utility for university ($M = 3.26$, $SD = 0.99$), utility for teaching ($M = 2.05$, $SD = 0.79$), and enjoyment of mathematics ($M = 2.40$, $SD = 0.79$). Of all participants, $n = 72$ (27.8%) were assigned to this cluster. This cluster is referred to as the "low-affinity" cluster, since participants indicated low agreement to appreciation of the mathematics discipline in all five variables.

Cluster 2. In this cluster, participants indicated values around the scale mean for intrinsic value ($M = 3.67$, $SD = 0.72$), personal importance ($M = 3.79$, $SD = 0.68$), utility for university ($M = 3.96$, $SD = 0.74$), utility for teaching ($M = 3.50$, $SD = 0.90$), and enjoyment of mathematics ($M = 3.67$, $SD = 0.61$). This cluster includes $n = 139$ (53.7%) participants. We refer to this one as the "moderate-affinity" cluster, as the indicated values were around the scale means.

Cluster 3. Participants in this cluster indicated a high intrinsic value ($M = 5.02$, $SD = 0.57$), personal importance ($M = 5.16$, $SD = 0.56$), utility for university ($M = 5.03$, $SD = 0.60$), utility for teaching ($M = 4.47$, $SD = 0.83$), and enjoyment of mathematics ($M = 5.01$, $SD = 0.44$). This cluster includes $n = 48$ (18.5%) participants. As participants in this cluster showed high appreciation of the mathematics discipline in the five facets, we refer to it as the "high-affinity" cluster.

Fig. 1 shows an overview of the three clusters regarding the five variables of appreciation of the mathematics discipline. The three graphs run roughly parallel except around the value for utility for university: In the low-affinity cluster, utility for university has the highest value of the five variables, while in the other two clusters, utility for university holds an average value compared to the other four variables.

4.3. Detailed description of clusters

4.3.1. Background variables

The three clusters did not differ in mean age of the participants ($F(2,195) = 1.41$, $p = .246$), gender distribution ($\chi^2(6) = 11.03$, $p = .088$), their semester of studying mathematics ($F(2,195) = 0.29$, $p = .750$), or their semester of studying the second subject ($F(2,195) = 0.28$, $p = .754$). There was a significant difference in participants' second subject between the three clusters ($\chi^2(6) = 21.65$, $p = .001$); in detail, the moderate-affinity cluster (23.9% science subject, 25.7% language subject, 35.4% social science subject, 15.0% other subject) differed from the high-affinity cluster (66.7% science subject, 9.1% language subject, 15.2% social science subject, 9.1% other subject) ($\chi^2(3) = 21.18$, $p < .001$). The clusters did not differ in their distribution of participants across the experimental groups ($\chi^2(6) = 8.41$, $p = .210$) or regarding the two factors ($\chi^2(2) = 0.90$, $p = .637$ and $\chi^2(2) = 2.77$, $p = .251$).

4.3.2. Views on the mathematics discipline

Clusters differed in how many participants stated a positive academic focus in the open statement ($\chi^2(2) = 55.13$, $p < .001$). The percentage of participants for whom a positive academic focus was coded in the low-affinity cluster (6.9%) was significantly lower than

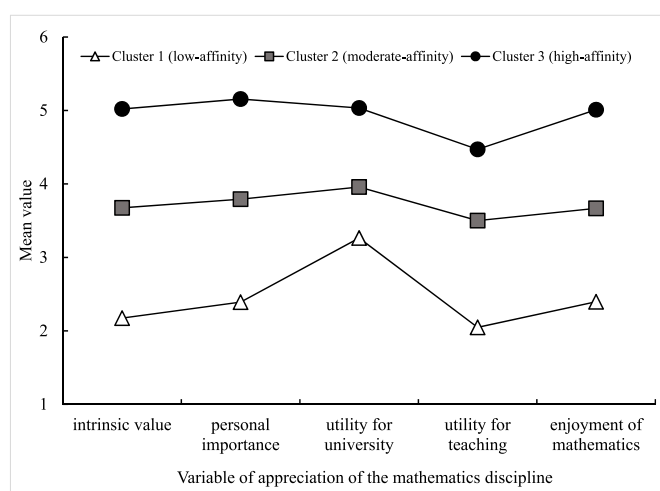

Fig. 1. Profile lines of the three clusters.

in the moderate-affinity cluster (41.0%; $\chi^2(1) = 26.52$, $p < .001$) and high-affinity cluster (72.9%; $\chi^2(1) = 56.41$, $p < .001$). In the high-affinity cluster, a positive academic focus was coded for significantly more participants than in the moderate-affinity cluster ($\chi^2(1) = 14.54$, $p < .001$). The clusters did not differ in regard to the number of participants for whom a negative academic focus was coded (low-affinity cluster: 2.8%, moderate-affinity cluster: 2.2%, high-affinity cluster: 2.1%; $\chi^2(2) = 0.10$, $p = .954$).

There was a significant difference between the clusters in how many participants mentioned a positive job-related focus ($\chi^2(2) = 9.02$, $p = .011$). In detail, the percentage of participants who mentioned a positive job-related focus was significantly lower in the low-affinity cluster (8.3%) than in the moderate-affinity cluster (21.6%; $\chi^2(1) = 5.88$, $p = .015$) and high-affinity cluster (29.2%, $\chi^2(1) = 9.00$, $p = .003$). The moderate-affinity and high-affinity clusters did not differ in regard to the number of participants for whom a positive job-related focus was coded ($\chi^2(1) = 1.14$, $p = .286$). A negative job-related focus for lacking relevance was coded for significantly different percentages per cluster ($\chi^2(2) = 19.39$, $p < .001$). The low-affinity cluster had a significantly greater percentage of participants (63.9%) who were coded as having a negative job-related focus for lacking relevance than did the moderate-affinity (47.5%; $\chi^2(1) = 5.13$, $p = .024$) and high-affinity clusters (22.9%; $\chi^2(1) = 19.39$, $p < .001$). In the moderate-affinity cluster, a negative job-related focus for lacking relevance was coded for significantly more participants than in the high-affinity cluster ($\chi^2(1) = 8.89$, $p = .003$).

5. Discussion

The aim of this study was to gain insight into preservice teachers' views on the discipline studied (i.e., mathematics) given the tension between an academic and a job-related focus on the discipline. Our initial plan to differently influence these views failed. Thus, using cluster analysis, we found that preservice teachers can be assigned to a cluster of either low, moderate or high affinity toward their study of the mathematics discipline. That is, preservice teachers who highly appreciated, moderately appreciated, or only slightly appreciated the mathematics discipline in regard to job-related facets of task value (i.e., utility for teaching) showed a similar appreciation in academic facets (i.e., enjoyment of mathematics or utility for university) and general facets (i.e., intrinsic value or personal importance). None of the clusters showed a high affinity in one of the facets but not another. However, preservice teachers who showed rather low affinity toward the mathematics discipline perceived a relatively high utility of the studied discipline for university compared to the other facets of appreciation. Consulting participants' open statements, low affinity toward the mathematics discipline corresponded to a less positive evaluation from both an academic and job-related focus and a more negative evaluation due to the content lacking relevance for one's future job compared to responses in the other two clusters. Analogously, a high affinity toward the mathematics discipline went along with more often viewing the mathematics discipline positively from an academic focus and less often viewing mathematics negatively from a job-related focus for lacking relevance. The clusters did not differ in background variables, except that participants who showed a high affinity toward the mathematics discipline were more likely to study another STEM discipline as a second subject than those who showed moderate affinity.

Although previous research has suggested that preservice teachers dominantly hold a job-related focus (Malmberg, 2006), neither the cluster structure nor the open statements supported this assumption. One reason for this might be the study setup, which aimed to be open for both an academic and a job-related focus. Nevertheless, the negative job-related focus for lacking relevance was used in all three clusters by a substantial share of participants (more than 20%). In addition, participants mentioned more often that relevance of the mathematics discipline for teaching was lacking than they would mention that there is substantial relevance of the mathematics discipline for teaching. When participants gave a positive evaluation of mathematics, this was primarily reasoned from an academic focus (e.g., interest in puzzling). Furthermore, participants in the low-affinity cluster mentioned mostly negative views, while participants in the moderate- and high-affinity clusters indicated sophisticated views that contained positive and negative aspects.

Common stereotypes exist for preservice teachers, specifically those who are strongly oriented only toward an academic *or* a job-related focus, but this study does not provide empirical evidence for existence of those stereotypes. Interestingly, preservice teachers who enjoy mathematics due to its rational character find the mathematics discipline also useful for future teaching. A focus on the academic nature of studying mathematics and a focus on the career goal of becoming a teacher seemed to be strongly connected in participants' views. For example, participants who showed a high affinity toward the mathematics discipline stated an interest in the proofs learned at university and how those can be taught at school. This is in line with prior research (Hannula, 2018) stating that ones' own experiences as a student at school shape the view that what is mainly important for teaching is the actual content taught, rather than associated background knowledge. However, in the low-affinity cluster, the academic focus needs to be considered in more detail: While participants in this cluster showed relatively low values in enjoyment of mathematics as a discipline, they perceived a relative high utility of the mathematics discipline for future university courses. This may be reasoned in the extrinsic nature of utility (Priniski et al., 2018). Furthermore, this difference between utility for university and utility for teaching highlights the importance of distinguishing not only the three main aspects of the construct of task value (intrinsic value, attainment value, utility value) but also what they refer to. Preservice teachers appear to ask themselves not only the question “Is studying mathematics useful?” but also “What is studying mathematics useful for?” (Alexander, 2018; Priniski et al., 2018).

Besides understanding preservice teachers' views on the mathematics discipline in terms of clusters, we need to discuss the initial idea of creating an intervention to prime an academic and/or a job-related focus for the discipline of mathematics. As outlined above, addressing an academic and/or job-related focus on the mathematics discipline had no main effects or interaction effects on the dependent variables, except for one small effect. Thus, we assumed that preservice teachers' appreciation of the mathematics discipline was rather a trait that could not be influenced by the short-term intervention administered (i.e., reading six quotes). This is supported by research on stability of task values during a mathematics lecture (Rach, 2023), although task values are perceived as situational in other learning contexts (Dietrich et al., 2017; Eccles & Wigfield, 2020). Further, while the quotes presented views on studying mathematics that we also partly found in participants' statements, each quote illustrated a view that included only one focus; thus, each quote represented a radical position. Therefore, participants might not have identified with the quotes and adopted the presented views.

5.1. Limitations

This study has some limitations. The choice of task value as a theoretical construct needs to be critically discussed. Although task value captures different facets of how a task can be appreciated,

these facets are strongly correlated (Wigfield & Eccles, 2020). In addition, many previous cluster and profile analyses have suggested rather similar values in all facets than characteristics in single facets (e.g., high utility value and low intrinsic value; Perez et al., 2014). Yet, the multifactor structure was supported by a confirmatory factor analysis in our sample, and we provided participants different focuses of task value (i.e., an academic and a job-related focus). Additionally, the subfacets of task value (especially utility for university) appeared to be not inseparable connected to each other. Thus, the differentiated conceptualization of appreciation in task value seems to be an acceptable framework.

Although this study incorporated mixed-methods data, the qualitative data could not be coded for all participants due to unelaborated answers. Many of these answers included strong feelings that could not be coded within the context of tension between an academic and a job-related focus. Because we could not consult participants for further questions, this study might lack some insights into participants' views.

5.1.1. Implications for research and practice

Those responsible for educating teachers can use the findings provided in this study to deal with preservice teachers. For example, lecturers could engage in a dialogue within courses to understand preservice teachers' views on the mathematics discipline, since background variables (e.g., semester) cannot be used as a predictor for how much affinity preservice teachers hold. Future research might focus on how the kind of affinity (low, moderate or high) is related to further variables, although many variables likely interrelate with appreciation, for example performance and study satisfaction (Kosiol et al., 2019; Perez et al., 2014; Schnettler et al., 2020). To precisely foster appreciation of the discipline preservice teachers have chosen to study, educators could take into account the view that the mathematics discipline is not relevant for teaching. Educators may want to address this view not only by mentioning how the discipline is relevant for teaching but also by describing why the mathematics discipline does not lack relevance for teaching, since many preservice teachers also hold a positive job-related view. Future research should investigate debunking such "myths".

Regarding the theory of task value, we successfully applied this framework to the context of preservice teachers' views on studying a discipline. Beyond for research in this context, the differentiation of utility value into utility for university and utility for teaching might be of interest for any research on task value: not only the kind of value (intrinsic, attainment, utility) may be crucial for learners' appreciation of the task but also what each kind of value refers to (Gaspard, Dicke, Flunger, Schreier, et al., 2015). For example, enhancing the utility of a task needs to be aligned to learners' views on that task (Canning & Harackiewicz, 2015). Further, definitions and measures of intrinsic value might be extended (Seo & Lee, 2018). As our study pointed out, enjoyment of mathematics appears to be an aspect of preservice teachers' appreciation of the mathematics discipline.

Further, for preservice teachers showing low affinity toward the mathematics discipline, appreciation may be particularly buoyed by focusing on utility for university. For example, teacher educators could first explain how the content and methods of a discipline are connected. Then, they could highlight how a single content area or the discipline as a whole can be useful for future teaching and underline that an understanding of the whole discipline is necessary for this.

Furthermore, and most importantly, the findings of this study support that preservice teachers' views on the mathematics discipline are stable and, therefore, cannot be influenced easily. Future research could focus on elaborated or more frequent interventions that might not only prime a certain view but instruct preservice teachers on to develop and reflect such views on their own (Belland et al., 2013; Schmidt et al., 2019).

5.2. Conclusion

All in all, this study shows that preservice mathematics teachers view the discipline studied from both an academic and a job-related focus. Thereby, preservice teachers have a broad range to view the mathematics discipline, and different levels of affinity occur in their views. The theoretical framework of task value proofed to not only predict motivation but – in our study – capture the broad range of how preservice teachers can appreciate the mathematics discipline between an academic and a job-related focus. Future research might apply and extend this framework. Further, teacher educators need to make the multidimensional views on scientific disciplines tangible in higher education teaching (Hillbrink & Jucks, 2019; Risan, 2022). From the first semesters on, even preservice teachers who show low affinity for their chosen discipline need to be introduced to the discipline from the perspective of a purely academic pursuit *and* as relevant for future teaching activities. Higher education might provide a space for reflecting ones' views and, thereby, might bridge the gap between theory and practice.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Preregistration

The preregistration for the initial research question can be found in the Open Science Framework at https://archive.org/details/osf-registrations-qnvpm-v1

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data is shared in the Open Science Framework https://doi.org/10.17605/OSF.IO/MJE3S

Acknowledgements

The authors are thankful to Celeste Brennecka for language editing and Johannes Illgner and Christopher Just help with qualitative analysis.

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